
Modern Bazaar

A Quantitative Market Intelligence System

Data Engineering, Market Metrics, Flipping, Risk, and Manipulation Ranking

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This document explains the implemented Modern Bazaar system in precise but readable terms. Equations describe the software as it exists; they are not promises of profit. Market conditions, hidden order-book depth, competing players, and game mechanics can cause realized outcomes to differ materially from model estimates.

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1 Executive Summary

Modern Bazaar is a market-intelligence platform for the Hypixel SkyBlock Bazaar. It turns frequent public API observations into a compact historical record, rolling liquidity metrics, risk estimates, and ranked trading opportunities. Its quantitative core answers two different questions:

1. **Flipping:** can a trader repeatedly buy at the current bid, sell at the current ask, and earn an attractive risk-adjusted profit within a chosen time horizon?
2. **Market manipulation analysis:** can the visible and estimated sell side be acquired within budget, and does historical order flow suggest that an inflated position could be exited before competing supply overwhelms it?

The system is deliberately empirical. Live prices describe the current opportunity; hourly summaries describe how the market normally behaves; rolling averages reduce noise; and scoring functions combine profitability with liquidity, competition, market control, attention, and adverse sell pressure.

The default manipulation model, called *Balanced*, is the most important recent development. It combines the bidder-attraction behavior of the Attention model with the sell-pressure discipline learned from the Overclocker model. Its design reflects a practical observation: attention is valuable only when a position can survive the sellers attracted by the same price movement.

2 System Boundary and Architecture

The production system contains five cooperating layers:

Layer	Responsibility
Hypixel API	Supplies current Bazaar products, prices, aggregate volumes, order counts, and the visible top of each order book.
Spring Boot core	Polls, validates, deduplicates, stores, compacts, aggregates, scores, and exposes REST endpoints.
PostgreSQL	Stores product metadata, raw snapshots, visible order-book entries, hourly summaries, retained hour points, and precomputed finance metrics.
Next.js dashboard	Presents searchable market data, strategy filters, ranked opportunities, and detailed plans.
Operations stack	Caddy terminates TLS and routes traffic; Prometheus collects application and database metrics; Grafana presents operational dashboards.

In production, the core services share an internal Docker network. Only Caddy binds public ports. It routes the API, Grafana, and Prometheus subdomains to their internal services. PostgreSQL is not directly exposed to the Internet.

2.1 Data-flow overview

The principal data path is:

Hypixel API $\longrightarrow$ snapshots $\longrightarrow$ hourly summaries and points
 $\longrightarrow$ rolling metrics $\longrightarrow$ strategy scores $\longrightarrow$ dashboard.

This separation is useful quantitatively. A live snapshot determines executable-looking prices and visible depth, while historical summaries determine whether those prices are supported by normal market behavior.

3 Market Notation

Hypixel’s API naming can be unintuitive, so this document uses economic notation.

Symbol	System field	Meaning
a_t	instant buy price	Lowest standing sell-order price: the approximate price paid to buy immediately. This is the market ask.
b_t	instant sell price	Highest standing buy-order price: the approximate price received when selling immediately. This is the market bid.
$s_t = a_t - b_t$	spread	Gross quoted spread per unit.
$\rho_t = s_t/b_t$	spread percentage	Spread relative to the approximate buy-order entry price.
V_t^B, V_t^S	buy/sell volume	Current aggregate unit depth reported by the API.
N_t^B, N_t^S	active buy/sell orders	Current number of standing orders on each side.
M_t^B, M_t^S	moving-week volumes	Rolling seven-day cumulative activity fields used to infer executed units between snapshots.
τ	tax rate	Fraction removed from sale proceeds; default $\tau = 0.01125$.

All prices are in coins and all flow metrics are normalized to units or events per hour.

4 Ingestion and Persistence

4.1 Polling and deduplication

The core polls the Bazaar endpoint every 60 seconds by default. If the API reports timestamp T , the system persists at most one snapshot for each pair

$$(\text{productId}, T). \quad (1)$$

This avoids storing repeated scheduler observations when Hypixel has not published a new market state. Products absent from the local catalog are inserted before their snapshots. Snapshots are written in batches of 50 to bound persistence-context memory.

For every product, a snapshot stores prices, reported volume and moving-week statistics, active order counts, weighted two-percent prices, and the visible buy and sell order entries returned by the API.

4.2 Hourly compaction

Every five minutes, a compaction job checks the oldest unprocessed snapshot. Once a complete one-hour interval plus a two-minute grace period exists, snapshots in that interval are reduced to an hourly summary and a selected set of representative points.

Let the snapshots in hour h be ordered as $x_0, x_1, \dots, x_n$. The summary records open, close, minimum, and maximum prices, plus order-flow and execution proxies. Representative points are retained when there is a sufficiently large time gap or price movement, preserving chart shape without retaining every raw observation forever.

The configured price-movement threshold is 0.2%, and the default maximum gap between kept points is five minutes. Conceptually, a point is retained when

$$\left| \frac{p_i - p_k}{p_k} \right| \geq 0.002 \quad \text{or} \quad t_i - t_k \geq 5 \text{ minutes}, \quad (2)$$

where k is the previously retained observation.

4.3 Retention

At application startup and daily at 03:00, the retention job deletes snapshots, hour points, and hourly summaries older than the configured retention interval. The repository documents a 45-day operational target, while the code-level default is 30 days if configuration is absent. Production behavior therefore depends on the active profile configuration.

The storage policy is a compromise:

- detailed recent observations support debugging and short-term analysis;
- hourly summaries preserve the variables needed by current scorers;
- representative hour points preserve visual price paths;
- bounded retention prevents indefinite database growth.

5 Hourly Market Metrics

5.1 Order-count activity proxies

For consecutive snapshots $i - 1$ and i , increases in active order count are treated as newly created orders:

$$C_h^B = \sum_i \max(0, N_i^B - N_{i-1}^B), \quad (3)$$

$$C_h^S = \sum_i \max(0, N_i^S - N_{i-1}^S). \quad (4)$$

These are *observed net-increase events*, not a perfect event log. If ten orders are created while ten existing orders disappear between polls, the active count does not change and those creations are unobservable.

When the order count rises, the implementation estimates newly added units using the absolute change in aggregate volume:

$$A_h^B = \sum_{i: N_i^B > N_{i-1}^B} |V_i^B - V_{i-1}^B|, \quad (5)$$

$$A_h^S = \sum_{i: N_i^S > N_{i-1}^S} |V_i^S - V_{i-1}^S|. \quad (6)$$

These variables are useful pressure proxies, but executions and cancellations occurring in the same polling interval can distort them.

5.2 Insta-trade flow

Positive changes in the rolling seven-day fields estimate executed flow:

$$I_h^{\text{sold}} = \sum_i \max(0, M_i^B - M_{i-1}^B), \quad (7)$$

$$I_h^{\text{bought}} = \sum_i \max(0, M_i^S - M_{i-1}^S). \quad (8)$$

I_h^{sold} represents units players insta-sold into standing buy orders. I_h^{bought} represents units players insta-bought from standing sell orders.

5.3 Top-bid chase behavior

The system measures how often the best bid rises:

$$U_h = \sum_i \mathbf{1}\{b_i > b_{i-1}\}, \quad (9)$$

$$\Delta_h^+ = \sum_i \max(0, b_i - b_{i-1}). \quad (10)$$

U_h is the number of upward moves and Δ_h^+ is their cumulative coin magnitude. The current Balanced formula uses U_h but exposes Δ_h^+ only diagnostically.

5.4 Rolling arithmetic averages

For metric X_h and a window containing n available hourly observations,

$$\bar{X}_n = \frac{1}{n} \sum_{j=1}^n X_{h-j+1}. \quad (11)$$

The system precomputes windows of 1, 6, and 48 hours every 60 minutes. If fewer than the requested number of observations exist, the mean uses the available observations. Thus the

stored `windowHours` may represent a target while the observation count represents the actual sample size.

Implementation note. The manipulation scorer currently uses the 48-hour averages. The flipping scorer uses all three windows and selects the smallest positive flow estimate, producing a conservative liquidity estimate.

6 Shared Price-Risk Model

The risk toolkit measures whether current prices are unusually far from two references: the latest weighted two-percent price and the historical mean closing price.

For value v and valid reference $r > 0$, define relative deviation

$$d(v, r) = \frac{|v - r|}{r}. \quad (12)$$

For each side of the book, the system takes the largest available deviation:

$$d_B = \max\{d(a_t, a_t^{2\%}), d(a_t, \bar{a}^{\text{close}})\}, \quad (13)$$

$$d_S = \max\{d(b_t, b_t^{2\%}), d(b_t, \bar{b}^{\text{close}})\}. \quad (14)$$

It also measures a reference shock between the weighted and historical values:

$$q = \max\{d(a_t^{2\%}, \bar{a}^{\text{close}}), d(b_t^{2\%}, \bar{b}^{\text{close}})\}. \quad (15)$$

The normalized risk score is

$$R = \min\left(1, \max\left(0, \max\left(\frac{\max(d_B, d_S)}{0.20}, \frac{q}{0.25}\right)\right)\right). \quad (16)$$

The market is flagged as likely atypical or already manipulated if

$$\max(d_B, d_S) \geq 0.15 \quad \vee \quad q \geq 0.12 \quad \vee \quad R \geq 0.85. \quad (17)$$

This is an anomaly heuristic, not a causal classifier: it detects unusual price regimes but cannot prove why they occurred.

7 Bazaar Flipping Model

7.1 Liquidity selection

Let D_w and S_w denote average insta-bought and insta-sold units for window $w \in \{1, 6, 48\}$ hours. The selected demand and supply are the smallest positive values:

$$D = \min^+(D_1, D_6, D_{48}), \quad S = \min^+(S_1, S_6, S_{48}). \quad (18)$$

This avoids projecting an unusually active short period across the whole trading horizon. Executable throughput is constrained by both sides:

$$Q = \min(D, S). \quad (19)$$

Markets with $Q < 1$ unit/hour receive score zero.

7.2 Budget, horizon, and competition

For budget B and horizon H , affordable units per hour are

$$Q_B = \left\lfloor \frac{B}{b_t H} \right\rfloor, \quad (20)$$

and planned capacity is $Q_P = \min(Q, Q_B)$ when a budget is supplied.

Let C_B and C_S be average created-order proxies. Define

$$C_{\max} = \max(C_B, C_S), \quad C_{\min} = \min(C_B, C_S), \quad (21)$$

$$\eta = \frac{C_{\max} - C_{\min}}{C_{\max} + 1}, \quad C_{\text{eff}} = C_{\max}(1 + 0.5\eta). \quad (22)$$

The side-balance adjustment is approximately

$$L = \min\left(1, \frac{S}{D + 1}\right), \quad (23)$$

with explicit edge handling for absent demand or supply. With competition coefficient κ , suggested units per hour become

$$Q_{\text{sugg}} = \frac{\min(Q_P H, Q H L)}{H \max(10^{-6}, 1 + \kappa C_{\text{eff}})}. \quad (24)$$

7.3 Profit and score

The current implementation treats the gross spread as the pre-risk unit margin and discounts it by risk:

$$\pi_u = s_t(1 - R), \quad (25)$$

$$\Pi_h = \pi_u Q_{\text{sugg}}. \quad (26)$$

Define

$$P = \log_{10}(\Pi_h + 1), \quad (27)$$

$$G = \min(0.8, 12\rho_t), \quad (28)$$

$$P_R = 0.1R, \quad (29)$$

$$P_C = \min(0.10, 0.0007C_{\text{eff}}), \quad (30)$$

$$L_Q = \min(1, Q/8). \quad (31)$$

The final flipping score is

$$\boxed{S_{\text{flip}} = P(1 + G)(1 - P_R)(1 - P_C)L_Q.} \quad (32)$$

Estimated fill times for the suggested quantity $Q_{\text{sugg}}H$ are

$$T_{\text{buy}} = \frac{Q_{\text{sugg}}H}{S}, \quad T_{\text{sell}} = \frac{Q_{\text{sugg}}H}{D}, \quad T_{\text{total}} = T_{\text{buy}} + T_{\text{sell}}. \quad (33)$$

Important: the implemented flipping margin does not explicitly subtract Bazaar tax. The score is therefore best interpreted as a comparative ranking signal, not an accounting statement of guaranteed net profit.

8 Manipulation Planning Model

This section documents the model used to rank markets that appear cornerable. It describes the system's analytical behavior; actual market actions remain exposed to hidden liquidity, competing supply, execution delay, game rules, and substantial loss risk.

8.1 Estimating total sell-side depth

The API exposes only the top portion of the order book. Let

- U_v be visible sell units,
- K_v be visible sell orders,
- K be the API's total active sell-order count,
- V^S be aggregate sell volume,
- C_v be cost of all visible units,
- p_{max} be the worst visible ask.

Missing orders are $K_m = \max(0, K - K_v)$. If $K_v > 0$, units are extrapolated as

$$\hat{U} = \max\left(U_v, U_v + K_m \frac{U_v}{K_v}, V^S\right). \quad (34)$$

The base hidden-unit price is conservatively selected as

$$p_h^{(0)} = \max\left(\frac{C_v}{U_v}, p_{\max}\right). \quad (35)$$

It receives a premium based on the proportion of hidden orders:

$$r_h = \min\left(1, \frac{K_m}{K_v}\right), \quad (36)$$

$$p_h = p_h^{(0)}(1 + 0.25r_h). \quad (37)$$

The estimated corner cost is therefore

$$\boxed{\hat{C} = C_v + \max(0, \hat{U} - U_v)p_h.} \quad (38)$$

This extrapolation deliberately charges hidden units at no less than the worst visible price. It cannot recover the true hidden distribution, but it is safer than assuming every hidden unit trades at the visible average.

8.2 Break-even, target, and projected profit

Average acquisition cost is

$$\bar{c} = \frac{\hat{C}}{\hat{U}}. \quad (39)$$

With sale tax τ , the break-even resell price is

$$p_{\text{BE}} = \frac{\bar{c}}{1 - \tau}. \quad (40)$$

For selected return multiplier $r \geq 1$ and sell-wall multiplier $w > 1$,

$$p_T = rp_{\text{BE}}, \quad (41)$$

$$p_W = wp_T. \quad (42)$$

The projected net profit per unit and total projected profit are

$$\pi_u = p_T(1 - \tau) - \bar{c}, \quad (43)$$

$$\Pi = \hat{U}\pi_u. \quad (44)$$

Because $p_T = r\bar{c}/(1 - \tau)$, this simplifies to

$$\Pi = \hat{C}(r - 1). \quad (45)$$

This identity is important: for fixed r , projected profit is proportional to capital deployed. The later quality terms must therefore prevent large but operationally poor markets from dominating the ranking.

8.3 Required bid movement and exit time

Starting from current best bid b_t , required doubling steps are

$$n_2 = \left\lceil \log_2 \left(\frac{p_T}{b_t} \right) \right\rceil_+, \quad (46)$$

where the value is zero when $p_T \leq b_t$.

Let $E = \bar{A}^B$ be average newly added buy-order units/hour. The estimated exit time is

$$T_E = \frac{\hat{U}}{E}. \quad (47)$$

Let A^S be added sell-order units/hour and I^{sold} be insta-sold units/hour. The adverse unit-pressure estimate is

$$P_S = A^S + I^{\text{sold}}. \quad (48)$$

The exit estimate assumes historical new buy-order units are potentially available to absorb the position. It does not guarantee that those orders appear above the required exit price or remain in place.

9 Common Manipulation Score

Before the versioned order-flow formula is applied, all manipulation models share the same hard gates and broad market-quality terms.

9.1 Hard gates

An opportunity receives score zero if any of the following holds:

$$\begin{array}{lll} \Pi \leq 0, & D < 1 \text{ unit/hour}, & E \leq 0, \\ n_2 > 6, & R \geq 0.85, & T_E > 24 \text{ hours}. \end{array}$$

The list endpoint adds practical filters for budget, maximum units, current item price, demand/-supply ratio, and minimum profit. Craftable and NPC-buyable products on maintained exclusion lists are removed. Higher-tier enchantments are removed when two lower-tier units can be combined more cheaply than the proposed target price.

With budget B , the default minimum-profit floor is

$$\Pi_{\min} = \min(0.10B, 50,000,000). \quad (49)$$

The default current item-price cap is 1.5 million coins.

9.2 Base quality factors

Projected profit is compressed logarithmically:

$$Q_{\Pi} = \log_{10}(\Pi + 1). \quad (50)$$

Corner-size quality favors operationally manageable inventories:

$$Q_U = \frac{10,000}{10,000 + \widehat{U}}. \quad (51)$$

Let D be insta-buy demand, S insta-sell supply, and

$$z = \begin{cases} \min(5, D/S), & S > 0, \\ 5, & S = 0, D > 0, \\ 0, & D = S = 0. \end{cases} \quad (52)$$

Demand quality is

$$Q_D = \frac{D}{D + 10} \left(0.35 + 0.65 \frac{z}{z + 3} \right). \quad (53)$$

Exit-speed quality is

$$Q_T = \frac{3}{3 + T_E}. \quad (54)$$

Bid-climb realism and anomaly survival are

$$Q_2 = \exp(-n_2/2.5), \quad (55)$$

$$Q_R = \min(1, \max(0, 1 - 0.9R)). \quad (56)$$

9.3 Standing market-control quality

Let active sell orders, active buy orders, sell units, and buy units be N^S, N^B, V^S, V^B . Define

$$t_{SO} = \frac{60}{60 + N^S}, \quad t_{SV} = \frac{50,000}{50,000 + V^S}, \quad (57)$$

$$d_{BO} = \frac{N^B}{N^B + 120}, \quad d_{BV} = \frac{V^B}{V^B + 50,000}, \quad (58)$$

$$i_O = \frac{N^B + 1}{N^B + N^S + 2}, \quad i_V = \frac{V^B + 1}{V^B + V^S + 2}. \quad (59)$$

Then

$$Q_M = 0.15 + 0.85(0.25t_{SO} + 0.25t_{SV} + 0.20d_{BO} + 0.15d_{BV} + 0.10i_O + 0.05i_V). \quad (60)$$

This rewards thin sell-side control and substantial buyer-side depth without allowing market control alone to reduce the score all the way to zero.

10 Order-Flow Signals

For readability, define the half-saturation transform

$$H(x; k) = \frac{x}{x + k} \quad (61)$$

and the squared quietness transform

$$L(x; k) = \frac{1}{1 + (x/k)^2}. \quad (62)$$

Using created buy orders C_B , created sell orders C_S , exit units E , adverse sell pressure P_S , bid raises U , and flipper-attention score F , the primitive signals are

$$B_H = H(C_B; 3) \quad \text{buy-order heat,} \quad (63)$$

$$S_Q = \frac{2}{2 + C_S} \quad \text{absolute sell-order quietness,} \quad (64)$$

$$B\% = \frac{C_B + 0.5}{C_B + C_S + 1} \quad \text{buy-side order share,} \quad (65)$$

$$P_Q = \frac{E}{E + P_S + 1} \quad \text{simple pressure quietness,} \quad (66)$$

$$C_Q = L\left(\frac{C_S}{C_B + 1}; 2\right) \quad \text{sell-creation dominance survival,} \quad (67)$$

$$P_Q^* = L\left(\frac{P_S}{E + 1}; 0.20\right) \quad \text{strong pressure survival,} \quad (68)$$

$$B_C = H(U; 3) \quad \text{bid-chase activity.} \quad (69)$$

10.1 Flipper attention

The attention proxy estimates whether the current quoted flip is attractive enough to be noticed. Let $Q_f = \min(D, S)$ and raw quoted flip profit/hour be

$$\Pi_f = (a_t - b_t)Q_f. \quad (70)$$

Then

$$F_\Pi = H(\Pi_f; 2,000,000), \quad (71)$$

$$F_\rho = H(\rho_t; 0.035), \quad (72)$$

$$F = \sqrt{F_\Pi F_\rho}. \quad (73)$$

This is an internal visibility proxy. It does not query external flipping websites and does not use the Modern Bazaar flipping rank directly.

11 Versioned Manipulation Formulas

The final manipulation score has the common form

$$S_{\text{manip}} = Q_{\Pi} Q_U Q_M Q_D Q_T Q_O Q_2 Q_R, \quad (74)$$

where Q_O is selected by formula version.

11.1 Overclocker formula (v1)

The original empirically successful model strongly rewards buy activity and penalizes adverse pressure:

$$Z_O = 0.30B_H + 0.20B_{\%} + 0.15S_Q + 0.25P_Q + 0.10C_Q, \quad (75)$$

$$Q_O^{(1)} = (0.05 + 0.95Z_O)(0.55 + 0.45C_Q)(0.10 + 0.90P_Q^*). \quad (76)$$

Its strength is practical market control. Its weakness is that a quiet, apparently favorable market may still fail to attract enough new bidders.

11.2 Attention formula (v2)

The Attention model adds direct bidder-chase and flipper-visibility requirements:

$$Z_A = 0.30B_H + 0.25B_C + 0.15B_{\%} + 0.15S_Q \quad (77)$$

$$+ 0.20P_Q + 0.05F + 0.05C_Q, \quad (78)$$

$$Q_O^{(2)} = (0.05 + 0.95Z_A)(0.35 + 0.65B_C)(0.45 + 0.55F) \quad (79)$$

$$\cdot (0.55 + 0.45C_Q)(0.10 + 0.90P_Q^*). \quad (80)$$

Its strength is evidence of visible competition. Its weakness is that requiring historical attention can favor markets where the crowd also supplies inventory aggressively.

11.3 Balanced formula (v3, default)

Balanced separates attraction from survival before recombining them. Define attention

$$A = 0.30B_H + 0.25B_C + 0.25F + 0.20B_{\%}, \quad (81)$$

pressure survival

$$G = 0.55P_Q^* + 0.25C_Q + 0.20P_Q, \quad (82)$$

and a broad base

$$Z_B = 0.25B_H + 0.20B_\% + 0.20P_Q + 0.15S_Q + 0.10B_C + 0.10F. \quad (83)$$

The order-flow quality is

$$Q_O^{(3)} = (0.05 + 0.95Z_B)(0.30 + 0.70A)(0.15 + 0.85G)(0.35 + 0.65P_Q^*)(0.50 + 0.50C_Q). \quad (84)$$

This is the key quantitative compromise. Attention is valuable, but it cannot compensate freely for sell pressure because pressure survival enters through several multiplicative terms. Conversely, a perfectly quiet market is not automatically ranked highly if it lacks buy heat, bid chasing, and flipper visibility.

12 Why Balanced Has Worked

Balanced corresponds closely to the actual execution problem:

1. A market must contain enough existing and newly created buyer capacity to provide an exit. This is represented by E , C_B , N^B , V^B , and B_C .
2. It must attract participants who perceive a profitable spread. This is represented by F and historical upward bid movement.
3. The position must remain controllable while those buyers arrive. This is represented by C_S , A^S , insta-sold units, standing sell depth, and corner size.
4. The price path must remain plausible and not already look anomalous. This is represented by n_2 and R .

The model is multiplicative because a fatal weakness should not be averaged away by an unrelated strength. For example, excellent projected profit should not rescue an item with nearly no exit flow. Balanced softens this principle with nonzero floors so that moderate imperfections reduce rather than automatically eliminate an opportunity.

13 Interpretation of Dashboard Outputs

Output	Correct interpretation
Corner cost	Estimated cost of visible plus extrapolated hidden sell-side units.
Average buy cost	Corner cost divided by estimated total corner units.
Minimum resell	Tax-adjusted break-even price, not a recommended guaranteed exit.
Projected profit	Profit if all cornered units exit at the chosen target multiplier.
Sell-through	Corner units divided by historical new buy-order units/hour. It is an optimistic capacity estimate, not a queue simulation.
Buy orders/hour	Average observed increases in active buy-order count. It may undercount true creation.
Sell orders/hour	Average observed increases in active sell-order count. Lower is generally preferable.

Output	Correct interpretation
Exit units/hour	Estimated units newly added to buy orders.
Pressure units/hour	Newly listed sell units plus units insta-sold into bids.
Bid raises/hour	Frequency of positive changes in the best bid, independent of jump magnitude.
Flip attention	Internal proxy derived from current spread and historical two-sided throughput.
Score	Relative ranking utility. It has no direct coin, probability, or expected-return unit.

14 Operational Observability

The application publishes Micrometer/Actuator metrics to Prometheus. Explicit ingestion counters include fetched products, persisted snapshots, and poll failures. PostgreSQL metrics are collected by a dedicated exporter. Grafana reads Prometheus as a provisioned data source.

Operational monitoring should answer four classes of questions:

- **Freshness:** is Hypixel being polled and are new snapshots arriving?
- **Completeness:** are hourly compaction and rolling aggregation keeping up?
- **Capacity:** are database size, connection usage, CPU, memory, and disk healthy?
- **Correctness symptoms:** did opportunity counts collapse, did metrics become flat, or did API error rates rise?

Prometheus is protected with HTTP Basic authentication at Caddy. Grafana uses its own login. The API and Actuator endpoints are protected by the application's security configuration.

15 Statistical and Modeling Limitations

15.1 Observation is not an event stream

The API provides periodic market states, not every order creation, cancellation, fill, or replacement. Metrics derived from state differences are therefore interval-censored. Events that cancel each other between polls cannot be reconstructed exactly.

15.2 Hidden depth is estimated

Only part of the order book is visible. The corner estimator extrapolates average units/order and applies a price premium, but a single large hidden order or sharply rising hidden price ladder can make actual acquisition cost much higher.

15.3 Averages mix market regimes

The 48-hour manipulation window stabilizes rankings but reacts slowly to sudden changes. Maintenance, game updates, weekends, creator attention, or an active manipulator can shift the current regime away from the historical mean.

15.4 Exit flow is not price-conditioned

E measures new buy-order units at historical prices. The current model does not estimate

$$\Pr(\text{new bid price} \geq p_T) \quad (85)$$

or model the probability that another seller consumes those units first. This is why displayed sell-through can be materially faster than realized liquidation.

15.5 Bid raises are sampling-sensitive

The count U_h depends on polling frequency and treats a minimal increase like a major move. The stored magnitude Δ_h^+ could eventually distinguish economically meaningful bidder competition from noise.

15.6 Scores are calibrated rankings, not probabilities

Neither strategy score is a probability of success. Coefficients and half-saturation constants encode practical preferences and observed behavior. Their strongest validation is comparative: better-ranked items should produce better realized outcomes on average under comparable execution.

16 Validation Framework

The next quantitative step should preserve the successful Balanced formula and evaluate it against a trade journal. For trade j , record at decision time:

$$x_j = (\hat{U}, \hat{C}, D, S, C_B, C_S, E, P_S, U, F, R, n_2, T_E, S_{\text{manip}})$$

and after execution:

$$y_j = (C_j^{\text{actual}}, T_j^{\text{exit}}, \Pi_j^{\text{realized}}, \text{drawdown}_j, \text{success}_j).$$

Recommended validation statistics include:

$$\text{Corner cost error}_j = \frac{C_j^{\text{actual}} - \hat{C}_j}{\hat{C}_j}, \quad (86)$$

$$\text{Exit-time error}_j = \frac{T_j^{\text{exit}} - T_{E,j}}{T_{E,j}}, \quad (87)$$

$$\text{Profit realization}_j = \frac{\Pi_j^{\text{realized}}}{\hat{\Pi}_j}. \quad (88)$$

Rank quality can be measured using Spearman correlation between score and realized profit, precision among the top k opportunities, median drawdown by score decile, and calibration curves for any later success-probability model.

An especially useful design is an immutable *decision snapshot*: store exactly the features and formula version shown when a trade is selected. This prevents later data updates from contaminating historical backtests.

17 Path Toward Simulation and Machine Learning

A future learning system should not begin by replacing the scorer. Balanced can serve as a strong policy prior and benchmark.

17.1 Replay environment

At decision time t , the state may contain live book features, rolling windows, risk, item metadata, and current inventory. An action specifies whether to enter, capital allocated, target price, wall configuration, and exit policy. Reward should be realized net profit minus explicit penalties for drawdown, time in position, and unliquidated inventory:

$$r_t = \Delta \text{coins}_t - \lambda_D \text{drawdown}_t - \lambda_T \text{capital time}_t - \lambda_I \text{inventory}_t. \quad (89)$$

The replay engine must enforce causality: the model may use only information available at time t . Random train/test splitting is inappropriate for market data; evaluation should use walk-forward time splits.

17.2 Recommended modeling sequence

1. Predict realized exit time and adverse sell pressure as supervised targets.
2. Predict probability of reaching several return thresholds within 1, 2, and 6 hours.
3. Use those calibrated predictions to rerank Balanced candidates.
4. Introduce policy optimization only after the replay environment reproduces actual historical trades with acceptable error.

This sequence is easier to debug than immediately training a reinforcement-learning agent and provides measurable improvements at every stage.

18 Recommended Quantitative Improvements

The following changes offer the highest information gain without discarding the empirically successful Balanced model:

1. **Outcome logging.** Record actual entry cost, fills, new orders, competing sells, exit time, realized profit, and formula version.
2. **Multi-window manipulation features.** Keep the 48-hour baseline, but expose 1-hour and 6-hour ratios to detect regime changes rather than replacing the baseline outright.
3. **Metric confidence.** Attach observation count, missing-hour ratio, and sample freshness to every rolling feature.

4. **Pressure decomposition.** Keep new sell listings and insta-sells separate in analysis so their independent effects can be estimated from outcomes.
5. **Price-conditioned exits.** Estimate the distribution of newly created bid prices, not only their total units.
6. **Magnitude-aware bidder chase.** Combine U_h with normalized Δ_h^+ while retaining the present count as a stable feature.
7. **Robust aggregation.** Compare arithmetic means with medians, trimmed means, and exponentially weighted averages during offline evaluation.

These are research directions, not immediate formula edits. The present Balanced model should remain versioned and reproducible while evidence is collected.

19 Conclusion

Modern Bazaar is best understood as a layered quantitative decision system. It does not merely sort products by spread. It observes the market repeatedly, compresses those observations into hourly behavior, compares current prices with historical references, estimates feasible trade size, and ranks opportunities using explicit economic constraints.

The Balanced manipulation formula is strong because it models the central tension directly: a profitable-looking market needs enough attention to create buyers, but not so much adverse supply that the position becomes uncontrollable. Its current success is plausible from both the mathematics and observed execution. The system's main uncertainties lie upstream in partial order-book visibility and state-difference proxies, and downstream in the absence of price-conditioned exit probabilities.

The correct next phase is measurement rather than uncontrolled coefficient tuning. A reliable decision-time feature log and realized-trade journal would transform the scorer from an empirically promising ranking into a system that can be calibrated, backtested, and eventually used as the foundation for simulation and machine learning.

A Principal Constants

Constant	Value	Interpretation
Default Bazaar sale tax	1.125%	Deducted from manipulation sale proceeds
Maximum demand/supply ratio	5.0	Prevents unbounded ratios near zero supply
Minimum demand	1 unit/h	Manipulation hard gate
Maximum exit time	24 h	Manipulation hard gate
Maximum bid doublings	6	Manipulation hard gate
Risk hard gate	0.85	Excludes highly atypical regimes
Demand half-saturation	10 units/h	Common demand quality
Exit-time half-saturation	3 h	Common exit quality
Created-buy half-saturation	3 orders/h	Order-flow buy heat
Bid-raise half-saturation	3 raises/h	Bidder-chase confidence
Sell-order half-saturation	2 orders/h	Sell quietness
Pressure-ratio half-saturation	0.20	Strong adverse-pressure penalty
Corner-size half-saturation	10,000 units	Practical inventory preference
Active sell-order half-saturation	60	Standing control
Active buy-order half-saturation	120	Standing buyer depth
Flip-attention profit half-saturation	2,000,000 coins/h	Internal visibility proxy
Flip-attention spread half-saturation	3.5%	Internal visibility proxy
Default max current item price	1,500,000 coins	List filter
Default automatic profit floor	10% of budget	Capped at 50 million coins

B Notation Summary

Symbol	Definition
a_t, b_t	Current ask and bid
D, S	Average insta-buy demand and insta-sell supply, units/hour
C_B, C_S	Observed created buy/sell order proxies, orders/hour
E	Newly added buy-order units/hour, used as exit capacity
P_S	Added sell-order units plus insta-sold units/hour
U	Positive best-bid movements/hour
F	Internal flipper-attention score in $[0, 1]$
R	Price-deviation risk score in $[0, 1]$
$\hat{U}, \hat{C}$	Estimated corner units and cost
p_{BE}, p_T, p_W	Break-even, target buy-order, and sell-wall prices
T_E	Estimated position exit time
$Q_O^{(1)}, Q_O^{(2)}, Q_O^{(3)}$	Overclocker, Attention, and Balanced order-flow quality